Project 2 Proposal: Predicting Unemployment

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Data resources

We plan to combine Google search activity and Reddit posts to estimate weekly initial unemployment insurance claims. Our proposed study period is October 2021 through September 2026, focusing on California, Washington, New York, and Texas.

Google Trends: We will collect state-level search interest from Google Trends for terms such as "file for unemployment" and "unemployment benefits." These values represent relative search interest rather than raw counts. We will use consistent download settings and check whether each state has sufficient weekly data.

Reddit: We will obtain posts through the Arctic Shift archive, using r/Unemployment for unemployment discussion and r/layoffs and r/cscareerquestions for tech-related discussion. We will aggregate posts by week and use explicit state mentions to construct state-specific measures where possible. These mentions identify the state being discussed, rather than necessarily the author's residence.

Unemployment claims: We will use weekly state-level initial unemployment insurance claims from the U.S. Department of Labor, also accessible through FRED. For example, CAICLAIMS provides weekly California claims without seasonal adjustment. We will account for seasonality and verify overlapping coverage across our sources before submitting.

Question

Do job-loss searches and Reddit discussions improve estimates of weekly state-level initial unemployment claims beyond what past claims already tell us?

Our project extends earlier Google-search nowcasting research to a recent period and tests whether Reddit adds useful information. The practical goal is to estimate the current week's claims before the official figures become available.

We will compare a baseline using past claims and seasonal controls with models that add search activity, Reddit activity, or both. We will examine predictions on later weeks excluded from model fitting and use only information available before each official release. This will establish whether online activity improves predictions beyond the baseline, rather than simply correlating with claims.

We will test whether tech-related Reddit discussion improves predictions more in California and Washington than in the other selected states. Starting in late 2021 excludes the exceptional 2020 claims surge from our main analysis. If online activity does not improve predictions, that result will still help establish the limits of these signals.